Problem 06: Counting Problem

Time limit: 1.5 seconds

You are given **N boxes**. The **i-th** box contains:

- $\mathbf{A_i}$ items of type **A**,
- $\mathbf{B_i}$ items of type **B**.

All items of type **A** are identical to each other.
All items of type **B** are identical to each other.

To form a sequence, you select **two different boxes**, take **all items** from both boxes, and then arrange all these items into **sequences**. You can do this operation with any two boxes.
The relative order of items in each sequence matters.

Two results are considered different if:

- The chosen pair of boxes is different, **or**
- The ordered sequences produced are different.

Compute the total number of different results, **modulo 1,000,000,007**.

Input

The first line contains an integer N $(2 \le N \le 200,000)$, the total number of boxes. N lines follow, each of which contain 2 space-separated integers $\mathbf{A_i}$ $(1 \le A_i \le 2000)$ and $\mathbf{B_i}$ $(1 \le B_i \le 2000)$.

Output

Print **one integer** — the total number of different results, **modulo 1,000,000,007**.

Sample Input	Sample Output
3 1 1 1 1 1 1	18

Explanation:

1 & 2 box: AABB, ABBA, ABAB, BBAA, BABA, ABBA

1 & 3 box: AABB, ABBA, ABAB, BBAA, BABA, ABBA

2 & 3 box: AABB, ABBA, ABAB, BBAA, BABA, ABBA

Total Count: 18